

Fake review detection system for online E-commerce platforms: A supervised general mixed probability approach

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ABSTRACT

Online consumer reviews play an important role in helping consumers judge the quality and authenticity of products on e-commerce platforms. However, the constant presence of fake reviews on these platforms has significantly impacted the operation and development of e-commerce platforms. In this study, we develop a novel supervised probabilistic method to detect fake reviews by utilizing the difference in the distribution of non-fraudulent reviews and that of fake reviews. Specifically, we first derive the univariate distributions of several unique features (linguistic, behavioral, and interrelationship features). We then integrate these distributions into two mixed distributions according to their labels to represent the overall difference between non-fraudulent reviews and fake reviews. Next, we randomly generate synthetic review data points with different labels from the above mixed distributions. Finally, we train a Multilayer Perceptron model by using these synthetic review data to obtain a classifier. We conducted several experiments to test the model using several original real-world review datasets. Numerical results indicated that the proposed supervised method outperformed some well-known sampling models and fake review detection methods, in terms of classification accuracy. Moreover, we extend the proposed method to handle the scenarios with small samples of raw review data. This study contributes to the literature by exploiting the difference in the distribution of non-fraudulent reviews and that of fraudulent reviews, which can improve the accuracy of fake review detection for online platforms.

1. Introduction

Growing rapidly over the last decade, e-commerce platforms have provided critical avenues for consumers to purchase products or services. According to a report by TIDIO, the number of worldwide online shoppers is approximately 2.14 billion, accounting for about 27% of the world's population [1]. As a result, a large number of user-generated online reviews constantly appear on e-commerce platforms. For example, YELP can generate up to 26,380 reviews per minute [2]. These reviews have provided consumers with useful information to enhance their understanding of products or services and, consequently, influence their purchasing behaviors [3–5].

However, fake reviews have appeared on these e-commerce platforms with the goal of influencing consumer perceptions of products and services and impacting sales [49]. These fake reviews can indeed affect consumer judgement of product quality, leading to reduced consumer trust and willingness to buy from e-commerce platforms [48]. Additionally, the proliferation of fake reviews gives rise to the economic

phenomenon known as “bad money expelling good money.” Specifically, poor-quality products increase sales rank with the help of fake reviews, causing high-quality products to gradually lose out in competition and ultimately resulting in significant declines in traffic and the reputation of e-commerce platforms [52]. Therefore, the development of technologies to automatically detect fake reviews plays a crucial role in the sustainability of these platforms.

Although fake reviews can vary and be well concealed, some features within the reviews can still be extracted by platforms to distinguish between fraudulent and non-fraudulent reviews. First, reviewers who post fake reviews have different motivations compared to non-fraudulent users, leading to their behaviors demonstrating statistical anomalies in comparison to non-fraudulent users [6]. For instance, fake review providers often write during a product's launch to dramatically increase or decrease its rating, whereas non-fraudulent reviewers post reviews in a random manner. Second, fake reviews differ significantly from non-fraudulent reviews in terms of their content or linguistic features. For instance, compared to non-fraudulent reviewers, fake

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reviewers tend to use an excessive number of emotive words, i.e., a higher emotional polarity, to express emotions within a review, subsequently influencing consumer decisions [50].

Previous studies have proposed various methods for detecting different types of fake reviews based on some behavioral features and linguistic features in recent years [21,47]. These methods employ several features to identify various types of fake reviews, such as reviews posted by users who consistently give the highest ratings to all products. However, several issues remain regarding effective fake review detection.

First, the content of a review or the behavior of writing a fake review can be easily replicated, making it challenging to accurately identify fake reviews solely based on these linguistic and behavioral features. Nevertheless, this does not imply that fake reviews lack suspicious clues. Certain characteristics become apparent when online manipulators, known as watermen, manipulate review ratings. Specifically, water armies aim to swiftly enhance the rating of their own product or lower the rating of a competitor's product, and build relationships with a larger number of users compared to regular reviewers. Therefore, it is crucial to consider the user-user relationship (interrelationship feature) in addition to linguistic and behavioral features. The interrelationship feature specifically refers to the relationship between reviewers and products in the bipartite graph, focusing on the associations between products, reviews, and reviewers. However, despite the construction of these features, few studies have considered the integration of the interrelationship feature into a fake review detection framework that already includes behavioral and content features. A new perspective that incorporates the interrelationship feature into the fake review detection framework will be beneficial for both practical application and research purposes.

Second, early studies have demonstrated significant differences between the distribution of fake reviews and that of non-fraudulent reviews, with a particular focus on product ratings. For instance, Dalvi et al. [8] found that user selection bias resulted in a heavily skewed distribution of product ratings across three domains (i.e., restaurants, movies, and products). More recently, a few studies have attempted to construct more comprehensive features that leverage the differences between the distribution of fake reviews and that of non-fraudulent reviews to improve detection accuracy [51]. For example, Kumar et al. [9] combined several heterogeneous feature distributions of review data (i.e., different facets of reviewer behaviors) into one mixed distribution. However, determining the appropriate probability threshold between fake and non-fraudulent reviews within the mixed distribution poses a challenge and can lead to a higher likelihood of misclassification. It is crucial to develop an innovative method based on probabilistic modeling to improve detection accuracy.

Lastly, recent studies have focused on addressing the issue of imbalanced real-world review datasets by using distribution-based balancing sampling algorithms. For example, Barushka et al. proposed an enhanced distribution-based balancing algorithm that generated new artificial spam samples [42]. They then trained a regularized deep multilayer perceptron neural network model on the synthetic dataset to detect spam. However, these methods assume that each feature follows a specific type of distribution (such as a Gaussian distribution), which is often not valid for real-world datasets, particularly when the dataset is small [9]. Therefore, it is necessary to calculate the natural distribution that each feature follows when developing a sampling algorithm based on probabilistic modeling, rather than relying on a predetermined distribution.

Overall, these three issues remaining from the prior literature motivate our study. To fill the gaps in the literature, this study first systematically integrates linguistic, behavioral, and interrelationship features of reviews, and then develops a novel probability-based approach to address the problem of the probability criterion for determining the boundary between fake reviews and non-fraudulent reviews. More precisely, we first transform the three review features into

corresponding optimal univariate sub-distributions using maximum likelihood estimation (MLE), rather than using a Gaussian distribution as the default. Then, we apply the Expectation–Maximization (EM) algorithm to derive the weight of each sub-distribution [43], and combine and sum all the sub-distributions to obtain two mixed distributions for two different classes of reviews. Finally, we utilize the synthetic review data generated by these two mixed distributions to train a Multilayer Perceptron (MLP) model to obtain the classifier, which can effectively identify fake reviews. It is worth noting that we detect fake reviews rather than fake reviewers by assuming that not every review written by a fake reviewer is fake.

The contributions of this study can be summarized as follows. First, this study systematically integrates linguistic, behavioral, and interrelationship features of reviews to build a comprehensive framework for fake review detection. We have not found any prior studies that have considered these three categories of features simultaneously in fake review detection. Integrating the three categories together, especially with the addition of interrelationship features, enables us to detect more comprehensive and suspicious clues of fake reviews, which have not been detected in the prior literature. Second and most importantly, we develop a novel supervised probabilistic approach to detect fake reviews by leveraging the distinguishing features between non-fraudulent and fake reviews. This study is one of the first to examine the differences in the distributions of the two types of reviews, to our best knowledge, which is a significant advance in the literature of fake review detection. It is important to note that the proposed supervised method not only significantly enhances the accuracy of fake review detection but also overcomes the limitations of the Stacking method proposed by Kumar et al. [9]. The results from a group of experiments indicate that the proposed method with several advanced detection methods is highly competitive in terms of classification accuracy, compared to other published methods in the prior literature. Third, the proposed method can be viewed as a new oversampling algorithm, as this study utilizes review data points generated from the mixed distribution to train the classifier and identify fake reviews. We also compare our method with several well-known sampling algorithms to evaluate the reliability of the synthetic review data generated from our method. The results confirm that our method can capture the structure of review data more effectively than the well-known sampling algorithms reported in prior studies.

The remainder of this study is organized as follows. Section 2 reviews several streams of literature. In Section 3, we introduce the supervised method. Section 4 reports several experiments that are conducted on three real-world datasets to investigate the performance of the proposed method. Finally, we draw conclusions and demonstrate future directions in Section 5.

2. Literature review

Our work is related to three streams of studies in the fake review detection literature. The first stream focuses on the types of fake review features used in detection methods, while the second stream examines various methods of probabilistic modeling. The final stream pertains to sampling algorithms used in the data preprocessing stage of fake review detection.

There are three types of review features used in the prior literature, i.e., linguistic features of reviews, behavioral features of reviewers, and hybrid features that include both linguistic and behavioral aspects. First, due to the diverse content of fake reviews, prior studies have incorporated linguistic features to improve the performance of fake review detection [10–12]. Second, other studies have considered behavioral features of reviewers to identify fake reviews [14–16] because reviewers tend to deliberately imitate non-fraudulent reviewers when composing their fake reviews. This makes it challenging to accurately identify fake reviews using only linguistic features. Therefore, several studies have shown that behavioral features contribute more to fake review detection

than linguistic features. Third, some studies have utilized hybrid features including both linguistic and behavioral aspects through feature fusion [18–20]. This approach has its advantage because it is difficult to comprehensively capture suspicious clues of fake reviews by separately utilizing linguistic and behavioral features.

While hybrid features can detect fake reviews, it is challenging to accurately identify professional-looking fake reviews that mimic the characteristics of real reviews. Therefore, the interrelationship feature has been considered for fake review detection, as it intuitively reflects real-world scenarios and exhibits good detection capabilities [7,22,24]. For example, Liu et al. integrated multimodal neural networks with attention mechanisms into probabilistic review graphs [23]. These graph-based methods have shown accurate detection of fake reviews. However, the applicability of these methods is limited because of their heavy reliance on labeled review data, as clear criteria for labeling fake reviews are currently lacking. Therefore, it is necessary to employ an unsupervised graph-based method. In this study, we utilize a graph-based fake review detection method proposed by Kawamoto et al. [25] as the interrelationship feature, as it does not require labeled data.

It should be noted that recent studies have continuously developed graph structures based on interrelationship features as a separate approach to detect fake reviews. However, few studies have explored the combination of the three different types of features, resulting in less information about the suspicious cues of fake reviews, which may affect the effectiveness of fake review detection. Therefore, we plan to combine these three types of features and build a more comprehensive framework for fake review detection, aiming to construct a broader set of suspicious clues for identifying fake reviews.

The second stream of literature related to our study focuses on the application of probabilistic modelling for fake reviews detection. Although various methods have been utilized in the literature, few studies have considered using probabilistic modeling [9]. Only a few approaches have considered utilizing rating distributions to detect fake reviews, as these approaches are not dependent on review content, which can significantly reduce the computational time required by the method [7,26–28]. However, a key limitation of these studies is that they only exploit the distribution of ratings as a feature and do not directly apply the differences between the distribution of fake reviews and that of non-fraudulent reviews. Consequently, there is a lack of examination regarding the differences in the distributions of other linguistic and behavioral features. More recently, Kumar et al. [9] introduced the concept of probabilistic modeling to translate behavioral features into corresponding distributions, aiming to reveal the difference between fake reviewers and non-fraudulent reviewers. The authors combined all the feature distributions into a mixed distribution using the EM algorithm.

Our proposed approach differs from Kumar et al. in three aspects [9]. First, our method focuses on detecting fake reviews rather than fake reviewers, which may reduce the risk of misidentification. The underlying assumption in detecting fake reviewers is that if a user is a fake reviewer, then all their reviews are fake. However, this assumption is not always valid as users may not always post fake reviews. Second, our method utilizes three different types of features, providing a more comprehensive perspective on the suspicious clues of fake reviews compared to solely relying on behavioral features of reviewers. Third, our method derives two mixed distributions using different labels for the review data, instead of a single mixed distribution. This approach can avoid the uncertainty associated with the probability of fake reviews in the mixed distribution. In summary, our proposed method introduces novel ideas for directly detecting fake reviews by leveraging various distributional features of reviews.

The third stream of literature is related to the sampling algorithm used in the data preprocessing stage of fake review detection. In real-world datasets, the number of fake reviews is typically much lower than that of non-fraudulent reviews. This imbalance causes the trained model to be biased towards non-fraudulent reviews, resulting in

inaccurate identification of fake reviews. To address this issue, several recent studies have used distribution-based balancing sampling algorithms. For example, Song et al. proposed a model called Conditional Anomaly Detection (CAD) that considers feature differences to detect conditional anomalies [44]. While these methods have demonstrated excellent performance across different datasets, they often assume that the distribution of each feature follows a Gaussian distribution, which is often not valid for real-world datasets, especially when the dataset is small [9]. In contrast, the current study calculates the natural distribution specific to each feature, including linguistic, behavioral, and interrelationship features, instead of relying on a default Gaussian distribution. Consequently, this study proposes a new oversampling algorithm as a complement to the existing literature.

In addition to the oversampling algorithm, a few studies have utilized the undersampling algorithm to address the imbalance in review datasets by removing a portion of the non-fraudulent reviews. For example, Rastogi et al. applied the random undersampling algorithm to preprocess the unbalanced YelpZip dataset and subsequently trained a support vector machine (SVM) model on the processed dataset [45]. However, a notable drawback of the undersampling algorithm is the removal of a significant number of non-fraudulent reviews, which may result in the loss of important features and potentially impact the classifier's performance. As a result, most studies that address the imbalance in fake review datasets tend to favor oversampling algorithms that not only retain important features but also increase the number of samples.

3. Methodology

In this section, we propose a Fake Review Detection method based on the General Mixed Probability Method, denoted as “FRDGMPM”. The overall design of FRDGMPM, as depicted in Fig. 1, consists of three components: feature extraction, mixed distribution construction, and classification model training. First, linguistic, behavioral, and interrelationship features are constructed to distinguish between non-fraudulent reviews and fraudulent reviews. To collectively extract these features, we apply the univariate distribution to fit the data in each feature. This approach helps generalize the constructed features since the distribution probability function can reduce the impacts of noise and outliers. Second, to address the highly imbalanced original dataset, we introduce a general mixed probability method (GMPM) to generate a balanced dataset. Specifically, we apply a Stacking method that incorporates multiple distributions of features to obtain a hybrid distribution using the EM algorithm. Moreover, the hybrid distribution constructed from the original dataset randomly generates synthetic review data points with different labels. Finally, a classification model is trained using synthetic review data to create a classifier, which is then tested for detecting fake reviews in the original real-life data. The detailed FRDGMPM is introduced in the remainder of this section.

3.1. Linguistic feature extraction

Reviewers fabricate fraudulent reviews with the aim of boosting or disparaging the reputation of a product or service, thereby influencing sales. This leads to a common phenomenon where fraudulent reviews

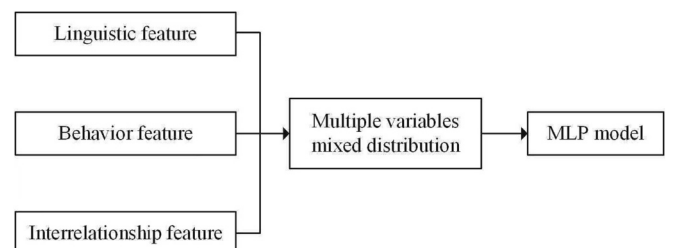


Fig. 1. The overall design of FRDGMPM.

often contain highly emotional statements [38], such as “Exactly as described by the seller, very satisfied, really like it”. Therefore, fraudulent reviews tend to include numerous adjectives and adverbs [17], resulting in a subjectively positive tone. This can create the false impression of a positive product experience for prospective consumers. Table 1 shows specific examples of reviews from the YelpChi dataset [40,41]. In summary, we propose several linguistic features that pertain to the content of reviews.

3.1.1. Basic linguistic feature

Reviewers often employ certain linguistic elements, such as repeated words, capitalized words, and emotional punctuation, to express their extreme opinions in fake reviews [17]. In addition, some reviewers write fake reviews that often include web links to boost the number of product searches and ultimately increase sales. Moreover, users tend to write fake reviews out of a haphazard mindset, which results in a shorter fraudulent review compared to a non-fraudulent review [31]. Consequently, the length of a review is an important feature of textual feature for identifying fake reviews. To distinguish between fraudulent and non-fraudulent reviews, we construct a series of fundamental linguistic features, i.e., repeated word, capitalized word, negative word, emotional punctuation (including exclamation mark, question mark, negative terms), web link, and text length [36,37].

3.1.2. Adjective/Adverb rate

Fake reviewers generally mimic the characteristics of non-fraudulent reviews when it comes to the details of their reviews, such as product descriptions and consumer experiences [39]. This explains why fake reviews contain numerous modifiers, such as adjectives and adverbs, to describe different aspects of the product. Therefore, the count of adjectives and adverbs in a review is constructed as an important linguistic feature for identifying fake reviews, specifically by utilizing part-of-speech (POS) tagging. Therefore, we divide reviews into a list of tagged words using the tag interface in the TextBlob toolkit [32,33], followed by counting the number of adjectives and adverbs.

3.1.3. Readability scores

E-commerce companies encourage consumers who have used their products to provide reviews. Product reviews on Amazon.com are marked with the number of people who found the review useful. In response to this practice, fake reviewers try to imitate non-fraudulent reviews and craft highly readable texts in order to gain consumer trust. In essence, fake reviews tend to exhibit higher readability

compared to non-fraudulent reviews, as they essentially imitate non-fraudulent reviews and are very intentional of enticing potential buyers. We use the Simple Measure of Gobbledygook (SMOG) [29,30] to measure the readability score of a review.

$$\text{SMOG} = 1.043 \sqrt{\text{number of polysyllables} \times \frac{30}{\text{number of sentences}}} + 3.1291. \quad (1)$$

3.1.4. Emotional polarity

Fake reviewers tend to post emotionally intense reviews in order to advocate or criticize products. For instance, there is a notable disparity in the distribution of sentiment words between Review 1 and Review 3 in Table 1. Review 3 is a non-fraudulent review, containing both positive and negative comments about different aspects of the hotel. In contrast, the fraudulent review (i.e., Review 1) includes many positive words with strong emotional intensity. Therefore, a fake review can be identified by calculating the sentiment score in the review to obtain sentiment polarity. Specifically, the sentiment score of a review is used as the sentiment feature, which can be derived using the sentiment interface in the TextBlob toolkit [32].

In Table 2, we provide a comprehensive list of specific linguistic features, along with their descriptions.

3.2. Behavioral feature extraction

Since fake reviewers deliberately mimic non-fraudulent reviewers when writing fake reviews, it becomes challenging to accurately identify fake reviews solely based on textual and linguistic features. For instance, Table 3 shows that five reviews received the votes of 100% usefulness from Amazon users due to limited fake clues, although they were proven to be fake reviews [41]. Although these reviews seem to be genuine, there are certain suspicious clues that can be found through an analysis of reviewer behavior. For example, the authors of these reviews are among the earliest reviewers of the respective products, and they have solely reviewed these specific products [41]. Therefore, we propose several features for identifying fake reviews. These features effectively depict the behavioral differences between fake reviewers and non-fraudulent reviewers.

3.2.1. Review count

Compared to normal users, fake reviewers often write many reviews to quickly enhance or disparage the reputation of a product. Therefore, we use the number of reviews posted by users to effectively identify this type of abnormal behavior [9].

3.2.2. Review gap

Fake reviews are often written at certain regular times to dramatically increase or decrease the rating of the product, whereas non-fraudulent reviewers post reviews randomly [40]. For instance, in an examination of the review gap between non-fraudulent reviews and fake reviews [31],

Table 1
Examples of types of reviews.

	Review Text	Type
1	This is an awesome hotel! My lovely daughter was married there in February. We had out of town guests staying there as well. Everyone raved about the hotel & the wedding ceremony & dinner afterwards. James Hotel did a fantastic job & it was a wonderful memory for all that were there!	Fraudulent
2	Get here early, myself and a friend waited an hour to be seated from 11–12 on a Sunday morning. Food was very fresh & original, I had the pouched eggs. Bloody Marys were awesome and I'm normally not a huge fan of that drink. Love the cozy feeling this place had. xaoI can, however, see people thinking that its a little too cramped, I give the waiters/waitresses so much credit for maneuvering their way through this room.	Fraudulent
3	I was skeptical staying at this hotel but I thought I should give it a shot...BIG. No. HUGE mistake. the room was ok but the paint at the wall was peeling. the bathroom has “curly” hair. someone had used the bathroom I checked in and forgot to flush. I had to inspect the bed for bed bugs ...too expensive and definitely not worth it. the management was very nice, accommodating and placed a wakeup call for me. nice people. the place appeared to be nice but turned out to be a disappointment.	Non-Fraudulent

Table 2
Linguistic features.

Feature Types	Feature	Description
Linguistic features	Repeated word	Total repetition words
	Capitalized word	The ratio of words with initial capital letter
	Negative word	Total negative words
	Exclamation mark	Total exclamation marks
	Question mark	Total question marks
	Weblink	Total weblinks
	Text length	Total words
	Adjective/ Adverb rate	The ratio of adjectives/adverbs
	Readability score	SMOG index
	Emotional polarity	Sentiment score

Table 3
Examples of fake reviews.

Reviewer	Product	Review content	Helpful rate
Big John	Audio Xtract(CD-ROM)	I can't believe for \$10 (after rebate) I got a program that gets me free unlimited music. I was hoping it did half what was...	1/1
Big John	Pond Aquarium 3D Deluxe Edition	This was a bargain at \$20 - better than the other ones that have no above water scenes. My kids get a kick out of the...	5/5
Cletus	Audio Xtract(CD-ROM)	This software really rocks. I can set the program to record music all day long and just let it go. I come home and my...	2/2
Cletus	Pond Aquarium 3D Deluxe Edition	This is one of the coolest screensavers I have ever seen, the fish move realistically, the environments look real, and the...	5/5
Jake	Pond Aquarium 3D Deluxe Edition	We have this set up on the PC at home and it looks GREAT The fish and the scenes are really neat. Friends and family...	3/3

a previous study showed that around 75% of fake reviewers posted more than 6 reviews per day, while more than 90% of non-fraudulent reviewers posted no more than 3 reviews per day. Therefore, this type of anomalous behavioral features of fake reviewers can be obtained by calculating the time difference between two adjacent reviews posted by the reviewers. The specific equation can be obtained as

$$RG_i = \frac{1}{n-1} \sum_{j=1}^n (T_{ij+1} - T_{ij}) \quad (2)$$

where RG_i is the average gap time of all the reviews posted by the i -th reviewer. T_{ij+1} is the time for the $(j+1)$ -th review written by the i -th reviewer, T_{ij} is the time for the j -th review written by the i -th reviewer, n is the total number of reviews written by the i -th reviewer.

3.2.3. Time difference

Fake reviewers tend to choose a specific time to post fake reviews. One common phenomenon is that a merchant typically hires fake reviewers at the beginning of launching a product to consistently write positive reviews for the product [52]. This is because a new product rarely has many reviews, consumers tend to peruse all available reviews, which may stimulate the sales for the product. Therefore, whether the reviews posted by users are early reviews reflects the abnormal behavior of these users. The specific calculation can be quantified as

$$TD_i = \frac{1}{n} \sum_{j=1}^n (T_{ij} - D_j) \quad (3)$$

where TD_i is the average time difference of the i -th reviewer, T_{ij} is the time for the j -th review posted by the i -th reviewer, D_j is the time of the initial review for the product posted by the j -th review. n is the total number of reviews posted by the i -th reviewer.

3.2.4. Extreme rating

The sheer volume of reviews for a product often makes it impractical for consumers to browse through all of them before making decisions. Fake reviewers take advantage of this by employing extreme ratings to manipulate the overall rating of a product. This distinctive behavior can be effectively identified using four distinct features. First, compared to regular users whose product ratings are randomly distributed across different scores, fake reviewers tend to concentrate their ratings heavily on extreme scores. Consequently, the entropy of product ratings from fake reviewers is lower than that of normal users. Second, fake reviewers

contribute a higher percentage of extreme reviews compared to genuine reviewers. Therefore, metrics such as the mean and variance of ratings given by users to all products, as well as the proportion of extreme ratings given by users to all their ratings, can be utilized to identify this anomalous behavior. Third, fake reviewers tend to assign higher ratings to a product compared to its average rating (rating deviation), which is another indicator of this abnormal behavior. In summary, we establish four extreme rating features, namely rating score, rating entropy, extreme rating rate, and rating deviation, to effectively identify and address this aberrant review behavior [9].

As shown in Table 4, we list the specific behavioral features of reviewers as well as their descriptions.

3.3. Interrelationship feature

Even though review content or the behavior of a fake reviewer can be easily mimicked, it is challenging to change the overall ratings of a product in a short period of time by one fake reviewer because there exist a lot of reviews for this product. Hence, some merchants always hire the Internet water army to manipulate reviews, which quickly boosts the rating of their products or reduces the ratings of the competitor's products [13]. Although it may be difficult to accurately identify a large number of fake reviews posted by the Internet water army, there is still some evidence of review manipulation which can be found through rigorous analysis of reviewers. For instance, because the impact of the product launch phase on consumer decision is relatively significant, the online water army usually publishes many reviews on a new product. However, it is difficult to use this anomalous behavior for detecting the online water army because early authentic reviews of a product are always few and far between, which makes it difficult to obtain useful information from these reviews [25]. Therefore, we apply a methodology based on the bipartite graph to characterize this spurious clue as the interrelationship feature. The graph can spread the impact of each reviewer's previous ratings and collect the impact of many rated products (<https://rgmining.github.io/ria/>) [25]. In this way, we can effectively address the two problems mentioned above and thus detect this type of fake reviewers.

3.4. General mixture distribution construction

All the content and behavioral features derived in the previous sections reflect suspicious clues that can be integrated into a model to accurately detect fake reviews. To address the problem of the probability criterion for determining the boundary between fake reviews and non-fraudulent reviews, by using a Stacking approach, this study develops a GMPM method to first incorporate the features of data points in each class as one unified mix probability distribution, respectively. Then, the synthetic review samples with different labels are generated from each uniformly mixed probability distribution derived above. In this way, our proposed GMPM method can combine all the textual and behavioral

Table 4
Behavioral features.

Feature Types	Feature	Description
Behavioral feature	Review count	Total reviews given by the reviewer
	Review gap	Mean/ S.D. of the time difference between two consecutive reviews
	Rating entropy	Entropy of ratings given by the reviewer
	Time difference	Mean/ S.D. of the difference between the time the reviewer reviewed the product and the time of the first review of the product
	Rating deviation	Mean/ S.D. of the ratings by the reviewer deviate from the average product ratings
	Extreme rating rate	The ratio of extreme ratings given by the reviewer
	Rating scores	Mean/S.D. of ratings given by the reviewer

features for providing comprehensive cues about fake reviews. Specifically, the GMPM model can utilize the machine learning technique to estimate its parameters during the training process, following three steps which include selecting the general mixed probability model, defining the objective function of the mixed distribution, and determining the optimal value of the sub-distribution weight. The overall process can be described in detail as follows.

3.4.1. Mathematical expression

The core of the FRDGMPPM method is to construct two unified mixed probability distributions for two sample classes with the aim to address the problem of the probability criterion for determining the boundary between fake reviews and non-fraudulent reviews. Specifically, the FRDGMPPM method first utilizes the original data to derive the corresponding optimal distributions of all the behavioral and content features for each class label. Then, the method determines the parameters and weights of all the sub-distributions in a mixture distribution with respect to each of these features using the EM algorithm [9]. Overall, the two unified mixed probability distributions are determined in two stages. First, all the behavioral and content features are first transformed into corresponding distributions using the maximum likelihood estimation (MLE). In other words, the parameters within each distribution are determined using the original data rather than following the EM algorithm. Second, the weight of each sub-distribution is determined by iteratively running the EM algorithm. Therefore, after deriving a sub-distribution for each content or behavioral feature using the raw review data, we focus on how to calculate the weight of each sub-distribution. The probability density function of the mixed distribution $f(x)$ is first given as

$$f(x) = \sum_{k=1}^K w_k f_k(x; a_k), \quad (4)$$

where $f_k(x; a_k)$ is the density function of the distribution corresponding to the k -th content or behavioral feature, a_k is the parameter of the k -th sub-distribution and let $\mathbf{a} \triangleq (a_1, \dots, a_K)$, w_k is the weight of the k -th sub-distribution and let $\mathbf{w} \triangleq (w_1, \dots, w_K)$, satisfying $\sum_{k=1}^K w_k = 1$. All the parameters of FRDGMPPM are denoted as $\theta \triangleq (\mathbf{w}, \mathbf{a})$.

3.4.2. Objective function

According to the probability density function of the mixed distribution, the probability density function of x_i is given as

$$f(x_i) = \sum_{k=1}^K w_k f_{ik}, \quad (5)$$

where $f(x_i)$ is the marginal probability distribution function of x_i , let $\mathbf{x} \triangleq (x_1, \dots, x_i, \dots, x_n)$ and $\mathbf{x}_i \triangleq (x_i^1, \dots, x_i^k, \dots, x_i^K)$. f_{ik} is the probability of the

k -th sub-distribution corresponding to the i -th review data point. Thus, the probability distribution of the observations $\mathbf{x}$ is given as

$$f(\mathbf{x}) = \prod_{i=1}^N f(\mathbf{x}_i) = \prod_{i=1}^N \sum_{k=1}^K w_k f_{ik}. \quad (6)$$

The objective function can be defined as

$$\ln f(\mathbf{x}) = \prod_{i=1}^N \ln \sum_{k=1}^K w_k f_{ik}. \quad (7)$$

3.4.3. Finding the optimal parameter

The general mixed model $f(\mathbf{x})$ mentioned above does not have an analytical solution, meaning that the optimal solution cannot be directly obtained using MLE. Therefore, following the procedures described in the literature [9], the weights of all the sub-distributions are determined using the EM algorithm. The EM algorithm provides an efficient iterative approach to optimize the estimation of parameters, including the following three steps:

- (1) Random initialization of the weight of each sub-distribution.
- (2) E step: the Q function can be constructed as

$$\mathcal{Q}(\theta^m, \theta) = \sum_{i=1}^N \sum_{k=1}^K \frac{w_k f_{ik}}{\sum_{j=1}^K w_j f_{ij}} [\ln w_k + \ln f_{ik}]. \quad (8)$$

- (3) M step: the equation for updating weight parameter θ^{m+1} after the $(m+1)$ -th iteration can be obtained

$$\theta^{m+1} = \underset{\theta}{\operatorname{argmax}} \mathcal{Q}(\theta, \theta_m). \quad (9)$$

where the Q function satisfies $\sum_{k=1}^K w_k = 1$.

$$\mathcal{Q}(\theta^m, \theta) = \sum_{i=1}^N \sum_{k=1}^K \frac{w_k f_{ik}}{\sum_{j=1}^K w_j f_{ij}} [\ln w_k + \ln f_{ik}] - \lambda \left(\sum_{k=1}^K w_k - 1 \right). \quad (10)$$

The final weight w_k is represented as

$$w_k = \frac{\sum_{i=1}^N \frac{w_k f_{ik}}{\sum_{j=1}^K w_j f_{ij}}}{N}. \quad (11)$$

The FRDGMPPM method is developed as shown in Algorithm 1.

Algorithm 1: Fake Review Detection Bases General Mixed Probability Method

Data: Review dataset R ; True reviews R_T ; Fake reviews R_F
Result: Synthetic samples x_T, x_F labeled as True or False

- 1 Compute the optimal univariate distribution P_T, P_F of R_T, R_F using MLE
- 2 Initialize weights $w_{T1} \dots w_{TK}, w_{F1} \dots w_{FK}$ randomly
- 3 **while** $w_{T1} \dots w_{TK}, w_{F1} \dots w_{FK}$ have converged **do**
- 4 Return $G_T = \sum_{k=1}^K w_{Tk} P_{Tk}, G_F = \sum_{k=1}^K w_{Fk} P_{Fk}$
- 5 **end**
- 6 **for** G_T, G_F **do**
- 7 Randomly generate initial review data from D_T, D_F ;
- 8 $x_T \triangleq (x_{T1}, \dots, x_{TK}), x_F \triangleq (x_{F1}, \dots, x_{FK})$ = Randomly generated initial review data
- 9 **end**
- 10 Compute the probability x_T, x_F belong to the mixed distribution G_T, G_F
- 11 $p(x_T) = \sum_{k=1}^K w_{Tk} f_{Tk}, p(x_F) = \sum_{k=1}^K w_{Fk} f_{Fk}$, using Eq. (5)
- 12 **if** $p < p(x_T) < 1, p < p(x_F) < 1$ **then**
- 13 Return Synthetic samples x_T, x_F
- 14 **end**

3.5. Classification model training

As a popular deep learning method, MLP has been widely utilized in various fields, including text classification, speech recognition, and fake review detection [46]. Recent studies have demonstrated that the MLP model provides more accurate fake review detection compared to well-known methods, particularly for Yelp datasets. For example, Budhi et al. proposed 133 unique features (including behavioral and linguistic features) to identify fake reviews using several well-known classifiers across three Yelp datasets. The experimental results showcased the superior performance of the MLP model in terms of classification accuracy, outperforming Logistic Regression (LR) and Linear Support Vector Machine (LSVM) [18]. Similarly, in this study, the MLP model is selected as the classifier for FRDGMPM.

4. Experiment

In this section, we introduce three real-world datasets and four evaluation measures (including accuracy, precision, recall, and F-measure) to evaluate the performance of FRDGMPM. To achieve accurate fake review detection, we conducted some experiments to determine the optimal number of synthetic samples generated by the mixed feature distribution for the proposed method. Based on the above parameter setting, three groups of comparative experiments were conducted to investigate the performance of FRDGMPM. It should be noted that, as the review data were randomly generated by FRDGMPM, the procedure to train the MLP model was repeated ten times in order to achieve statistical significance. Then, the average of these ten predicted results was recorded as the final result of each tested method in this study. In addition, we tested the proposed method with a small amount of labeled fake review data points to demonstrate the practical application of the method in real scenarios. All numerical experiments were performed using the python PyCharm platform on a personal desktop.

4.1. Datasets and features

We utilized three real-world review datasets (YelpZip, YelpNYC, and YelpChi) to validate the performance of FRDGMPM. Specially, these datasets are collected by Yelp.com from non-fraudulent consumers on its platforms, which has been proven to be highly approximate to real-world fake reviews [34]. The YelpZip dataset is the largest of the three datasets, including 608598 reviews of all relevant restaurants in the states of New York, New Jersey, Vermont, Connecticut, and Pennsylvania. In the YelpNYC and YelpChi datasets, there are 322167 reviews about 5044 restaurants and 5854 reviews about several hotels located in the New York City and Chicago area, respectively. Based on the original datasets, we constructed 22 features that are defined in Section 3.

4.2. Evaluation measures

We used the following measures to evaluate the performance of the

proposed method. True positive (TP) is defined as the number of reviews in which FRDGMPM correctly identifies fake reviews, while fake positive (FP) is defined as the number of reviews in which FRDGMPM identifies fake reviews as non-fraudulent reviews. Similarly, false negative (FN) is defined as the number of reviews in which the FRDGMPM model incorrectly identifies non-fraudulent reviews as fake reviews, while true negative (TN) is correctly defined as the number of reviews in which the prediction model identifies non-fraudulent reviews as non-fraudulent reviews [17]. Then, four common evaluation measures that are used in the prior literature, i.e., accuracy, precision, recall, and F1 score, are calculated by using the evaluation criteria of the MLP model.

4.3. Parameter settings

In the process of weight learning for FRDGMPM, the size of the weight and the critical value impacted the convergence speed. We empirically determined the threshold value for each sub-distribution iteration as 0.001. In other words, when the weight difference of the sub distributions in the mixed distribution was less than 0.001, the EM algorithm stopped iterating and then generated the final weight of the sub-distribution.

To select the optimal MLP network structure, we first empirically set the range of the number of hidden layers (N_{hl}) from 1 to 10 and the range of the number of neurons in each hidden layer (N_{ne}) from 1 to 10. Next, we determined the optimal values for N_{hl} and N_{ne} comparing the detection accuracy of MLP when different combinations (N_{hl}, N_{ne}) were selected in the three datasets. It should be noted that we calculated the sum of accuracy and F-measure as a new metric to demonstrate the impact of different network structures on fake review detection. Fig. 2 shows that FRDGMPM achieved higher accuracy when (N_{hl}, N_{ne}) was set to (3, 2) for the three datasets. Therefore, to achieve good performance, we set (N_{hl}, N_{ne}) as (3, 2) for the rest of the numerical experiments.

4.4. Impact of the amount of synthetic review data generated by FRDGMPM

As discussed in Algorithm 1, the amount of synthetic review data generated by the mixed distribution N affects the detection performance of FRDGMPM. Specifically, a small N may lead to poor performance because it is often difficult to train the MLP model with higher accuracy in a smaller amount of data. In contrast, a large N can result in high computational cost, which affects the efficiency of the MLP model detection. Hence, it is necessary to conduct experiments with the aim of finding the optimal N^* to balance efficiency and accuracy. According to the principle of the FRDGMPM method described in Algorithm 1, N is affected by two variables, i.e., the number of initial synthetic review samples $N_{initial}$ and the probability of obeying the mixed distribution p . Specially, in order to control the range of N to find the optimal N^* , we first empirically set $N_{initial}$ as 600 (selecting from 100, 200, ..., 1000) and p as 0.2 (selecting from 0.05, 0.1, ..., 0.5). Second, the optimal N^* determined by comparing the detection accuracy of MLP when different N

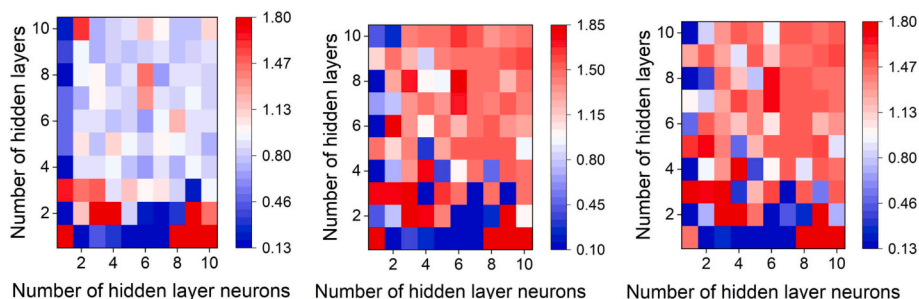


Fig. 2. Accuracy of FRDGMPM on three datasets with different combinations (N_{hl}, N_{ne}).

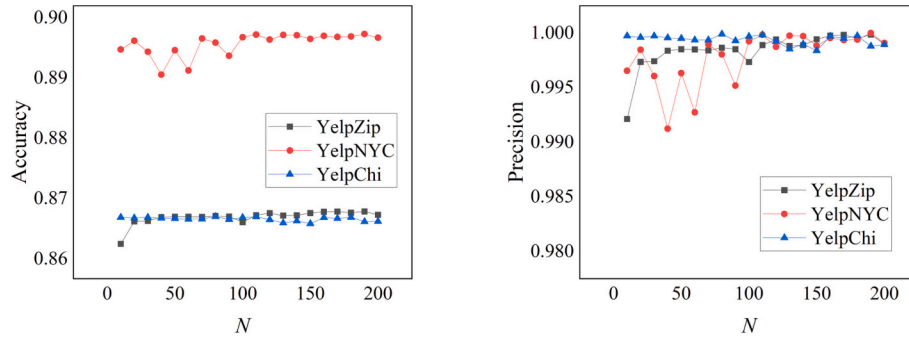


Fig. 3. Accuracy and Precision of FRDGMPM on three datasets with different N .

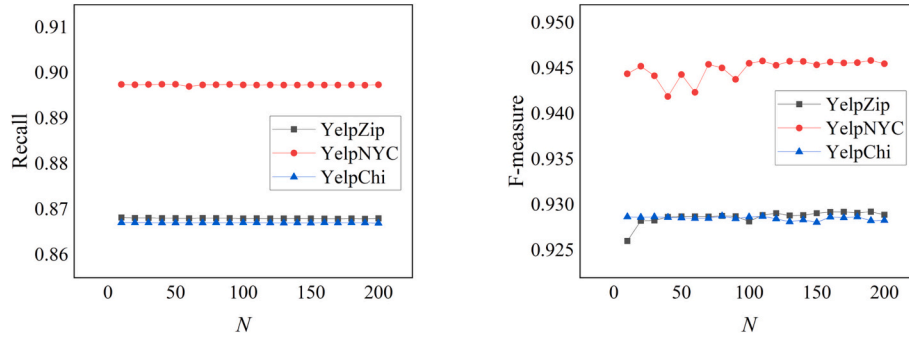


Fig. 4. Recall and F-Measure of FRDGMPM on three datasets with different N .

values are selected. In addition, for the sake of simplicity, we set the ratio of the number of fake reviews to that of non-fraudulent reviews as 1:1 in this study. Figs. 3 and 4 show the results from using different values of N on YelpZip, YelpChi, and YelpNYC, respectively. Fig. 3 shows that FRDGMPM produces more accurate detections of fake reviews when N is 190 on three datasets. Although the proposed method achieves better recall when N is 50 using YelpNYC in Fig. 4, the F-Measure, which is a compromise between precision and recall, still reaches the optimal value when N is 190 using YelpNYC. Therefore, to achieve good performance, we set N as 190 for the rest of the numerical experiments.

4.5. Model comparison and validation

We conducted three groups of comparative experiments to investigate the performance of FRDGMPM. In fact, the core of the proposed method is to train a classifier by generating synthetic review data points to identify fake reviews. It is necessary to evaluate the quality of the synthetic samples generated by the mixed distributions, as this is directly related to the accuracy of the proposed method for detecting fake reviews. Therefore, in the first group of experiments, we utilized

Table 5
Summary of Experiment 1

Dataset	Sampling algorithm	Acc (%)	Pre (%)	Rec (%)	F1 (%)
YelpZip	SMOTE	68.68	96.40	66.41	78.61
	RandomOverSampler	68.94	96.33	66.79	78.81
	ADASYN	69.55	96.15	67.66	79.36
	(Ours)	86.79	99.99	86.80	92.92
YelpNYC	SMOTE	68.81	96.70	67.57	79.52
	RandomOverSampler	67.37	97.09	65.62	78.27
	ADASYN	67.42	97.04	65.72	78.33
	(Ours)	89.73	100.00	89.73	94.59
YelpCHI	SMOTE	16.10	99.65	3.26	6.28
	RandomOverSampler	15.78	99.80	2.88	5.59
	ADASYN	27.92	97.56	19.44	20.82
	(Ours)	86.62	99.88	86.70	92.83

Table 6
Summary of Experiment 2.

Dataset	Description	Acc (%)	Pre (%)	Rec (%)	F1 (%)
YelpZip	linguistic	13.29	0.16	70.63	0.32
	behavior	79.48	87.16	89.00	88.03
	hybrid	86.79	99.99	86.80	92.92
YelpNYC	linguistic	42.13	40.13	80.79	38.11
	behavior	80.59	86.19	91.68	88.84
	hybrid	89.73	100.00	89.73	94.59
YelpChi	linguistic	13.37	0.10	96.26	0.20
	behavior	59.56	61.58	88.23	72.50
	hybrid	86.62	99.88	86.70	92.83

SMOTE, random oversampling, and ADASYN as the benchmark algorithms because these algorithms solve the class imbalance problem in the dataset by generating synthetic sample points [35].

Table 5 shows the performance of FRDGMPM and benchmark algorithms on three datasets. FRDGMPM produced more accurate detection than each benchmark algorithm in terms of accuracy, precision, recall, and F-Measure. This empirically demonstrates that the proposed method could more effectively capture the structure of fake and non-fraudulent reviews in the dataset by constructing two mixed distributions with different labels compared to benchmark algorithms.

In the second group of experiments, we applied each of the three different types of features as the input features to FRDGMPM to compare their individual contributions to fake review detection. As shown in Table 6, the proposed method significantly improved the accuracy of fake review detection compared to those using either linguistic features or behavioral features. This improvement is due to the hybrid features incorporating more comprehensive information regarding suspicious clues of fake reviews. It is worth noting that better results were obtained using behavioral features instead of linguistic features in YelpZip and YelpNYC, indicating that behavioral features were relatively more important than linguistic features for fake review detection.

Moreover, the contribution of the interrelationship feature to fake

Table 7

Summary of experiment 2: with/without interrelationship feature.

	YelpZip		YelpNYC		YelpChi	
	Full feature	Without interrelationship feature	Full feature	Without interrelationship feature	Full feature	Without interrelationship feature
Accuracy (%)	86.79	60.15	89.73	76.97	86.62	63.82
Precision (%)	99.99	61.22	100.00	80.63	99.88	67.83
Recall (%)	86.80	86.88	89.73	92.75	86.70	87.67
F-Measure (%)	92.92	67.85	94.59	86.26	92.83	76.39

Table 8

Summary of Experiment 3.

Dataset	Description	Acc (%)	Pre (%)	Rec (%)	F1 (%)
YelpZip	NB [20]	–	–	90.04	70.2
	CNN [18]	77.78	28.86	88.94	43.58
	(Ours)	86.79	99.99	86.80	92.92
YelpNYC	MLP [20]	–	–	81.86	79.74
	CNN [18]	76.81	22.32	89.04	35.69
	(Ours)	89.73	100.00	89.73	94.59
YelpChi	SVM + BiGAN [15]	83	81.2	85.7	83.4
	RE + PE + Bigram [11]	86.5	84.2	89.9	87
	CNN + AFDA [19]	80	83.9	74.2	78.7
	CNN [18]	99.93	99.9	99.8	99.85
	(Ours)	86.62	99.88	86.70	92.83

review detection is demonstrated by comparing the detection accuracies of the proposed method using feature sets with or without interrelationship features, as shown in Table 7. The results indicated that the proposed method achieved more accurate detections using the full feature set compared to when the interrelationship feature was not used, across the three Yelp datasets. This empirical evidence demonstrated that incorporating the interrelationship feature could lead to more accurate fake review detection.

In the third group of experiments, following the procedures in the literature [21], we compared the proposed method with some state-of-the-art methods of fake review detection to validate the performance of FRDGMPPM. Table 8 shows the detection performance of FRDGMPPM and other advanced methods on three datasets. The results indicated that the proposed method produced more accurate detections than other baseline methods in terms of accuracy and F-Measure, especially for YelpZip and YelpNYC. Although the accuracy and F-Measure of FRDGMPPM were worse than the best method CNN in YelpChi [18], the results were significantly better in the other two datasets. Notice that YelpZip and YelpNYC had almost 100 times the number of reviews in YelpChi, which indicated that the proposed method was highly competitive with CNN in the scenario of big data.

4.6. FRDGMPPM method extensions

The core step of the proposed method described above is to use all the original review data to derive a mixed distribution with different

labels. Although this practice has been shown to outperform the baseline methods in the above experiments, it is usually difficult to obtain the labels of fake reviews in real situations. This requires us to examine whether we can get better detection performance using a small amount of review data. For this purpose, we implemented our idea by reducing the number of raw review data N_{raw} inputted into FRDGMPPM. In other words, we utilized only a small fraction of the raw review data as input to the proposed methods, which not only reduced the method's reliance on fake review data but also saved considerable computing time. Similarly, we still chose the original review data for the MLP model and the parameter settings remained the same as the experiments above. Figs. 5 and 6 show the results using different N_{raw} on the three datasets. Fig. 5 shows that FRDGMPPM achieved good accuracy and precision when N_{raw} was 100 for the three datasets. Similarly, Fig. 6 shows that our method produced more accurate detections when N_{raw} was 100 for the three datasets. Therefore, the proposed method could effectively identify fake reviews using only 100 raw review data, which means that the proposed method was close to semi-supervised learning.

5. Conclusion

In this study, we have proposed a novel supervised FRDGMPPM method by using the approach of probabilistic modeling to effectively detect fake reviews. Unlike prior studies [34,47], the proposed method mainly focuses on how to directly utilize the difference in distribution features between non-fraudulent and fake reviews, rather than using numerical features. Specially, being constructed with sub-distributions of novel features that include linguistic, behavioral, and interrelationship features, FRDGMPPM is designed to derive two mixed distributions to represent the overall difference between non-fraudulent and fake reviews. Then, we detect fake reviews by using the MLP model that is trained with synthetic review data generated from the above mixed distribution. Some numerical experiments were conducted on three real-world datasets to validate the performance of FRDGMPPM. Moreover, by reducing the amount of raw review data used in experiments, FRDGMPPM is shown to handle the cases with small samples well so that this method can be easily extended to semi-supervised learning. Major findings are summarized as follows.

First, this study extends the probabilistic approach proposed by Kumar et al. [9] for the first time, which can effectively solve the problem of unknown probability criteria for the boundary between fake

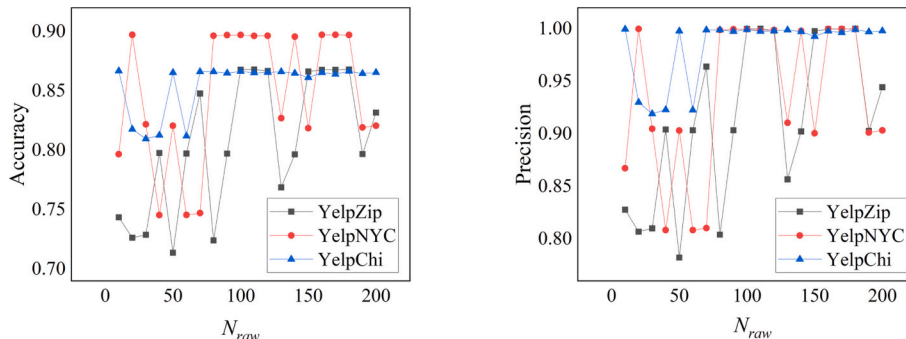


Fig. 5. Accuracy and Precision of FRDGMPPM on three datasets with different N_{raw} .

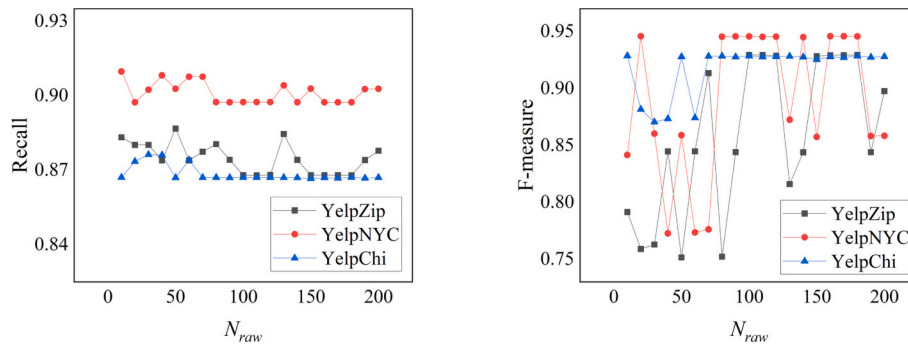


Fig. 6. Recall and F-Measure of FRDGMPM on three datasets with different N_{raw} .

and non-fraudulent reviews in mixed distributions. Moreover, numerical results confirmed that FRDGMPM also captured the structure of review data in the dataset more effectively than other well-known sampling algorithms. This innovative sample generation approach can be extended to other classification problems using small data samples with labels (such as cancer case prediction).

Second, prior studies have shown that behavioral features contribute more to fake review detection than linguistic features [17,18]. Similarly, our comparative experiments confirmed this view. Furthermore, we verified the contribution of the interrelationship feature to fake review detection, and the results showed that FRDGMPM using the full feature sets achieved much more accurate detections than using feature set without interrelationship feature. This finding inspires us to focus more on constructing interrelationship features in future studies.

Third, FRDGMPM outperformed some well-known state-of-art methods, indicating that our method can help consumers or e-commerce platforms identify fake reviews more effectively. Moreover, the empirical results are based on two big data sets, suggesting that FRDGMPM can be applied in the context of big data.

Our study provides several practical implications for e-commerce. Our findings suggested that behavioral features contributed more to fake review detection than linguistic features, indicating that platform managers should pay more attention to user behaviors. Meanwhile, interrelationship features are also important so that platforms should encourage reviewers to engage in communication with each other to provide rich interrelationship features. In addition, our method can be integrated into decision support systems to help platforms quickly and accurately determine whether a review should be deleted [21,47]. The extended FRDGMPM can achieve accurate detections with a small amount of original review data, which can be easily deployed by e-commerce platforms. While our study focuses on detecting fake reviews, our proposed method can be extended to detect fake bots within e-commerce platforms because the features we utilized in our study are very beneficial for identifying bots as well. In summary, the proposed method is generic, allowing platform managers to incorporate new features of fake reviews into our framework by computing the sub-distributions of these new features. This adaptability ensures that the framework can evolve with the emergence of new indicators for identifying fake reviews.

This study has identified several limitations that could be addressed in future research. First, while the proposed method theoretically supports a mixture of univariate and multivariate distributions, our study only considers univariate features, such as linguistic, behavioral, and interrelationship aspects, and does not model multivariate features for certain indicators of fake reviews. Therefore, future studies can explore the inclusion of multivariate features to enhance the detection framework. Second, one of the contributions of this study is the incorporation of the interrelationship feature into the fake review detection framework. However, we have only utilized one of several possible interrelationship features. This approach may result in an incomplete

collection of suspicious cues regarding fake reviews, potentially impacting the detection performance. To overcome this limitation, future research can expand on the integration of additional behavioral, linguistic, and interrelationship features into the proposed method. Examples include incorporating interdependent behavioral features and multimodal neural network-based interrelationship features.

CRedit authorship contribution statement

Jiwei Luo: Methodology, Formal-analysis, Data-curation, Writing-original-draft. **Jian Luo:** Conceptualization, Methodology, Supervision, Writing-review-editing. **Guofang Nan:** Conceptualization, Methodology, Supervision, Writing-review-editing. **Dahui Li:** Supervision, Writing-review-editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The authors do not have permission to share data.

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